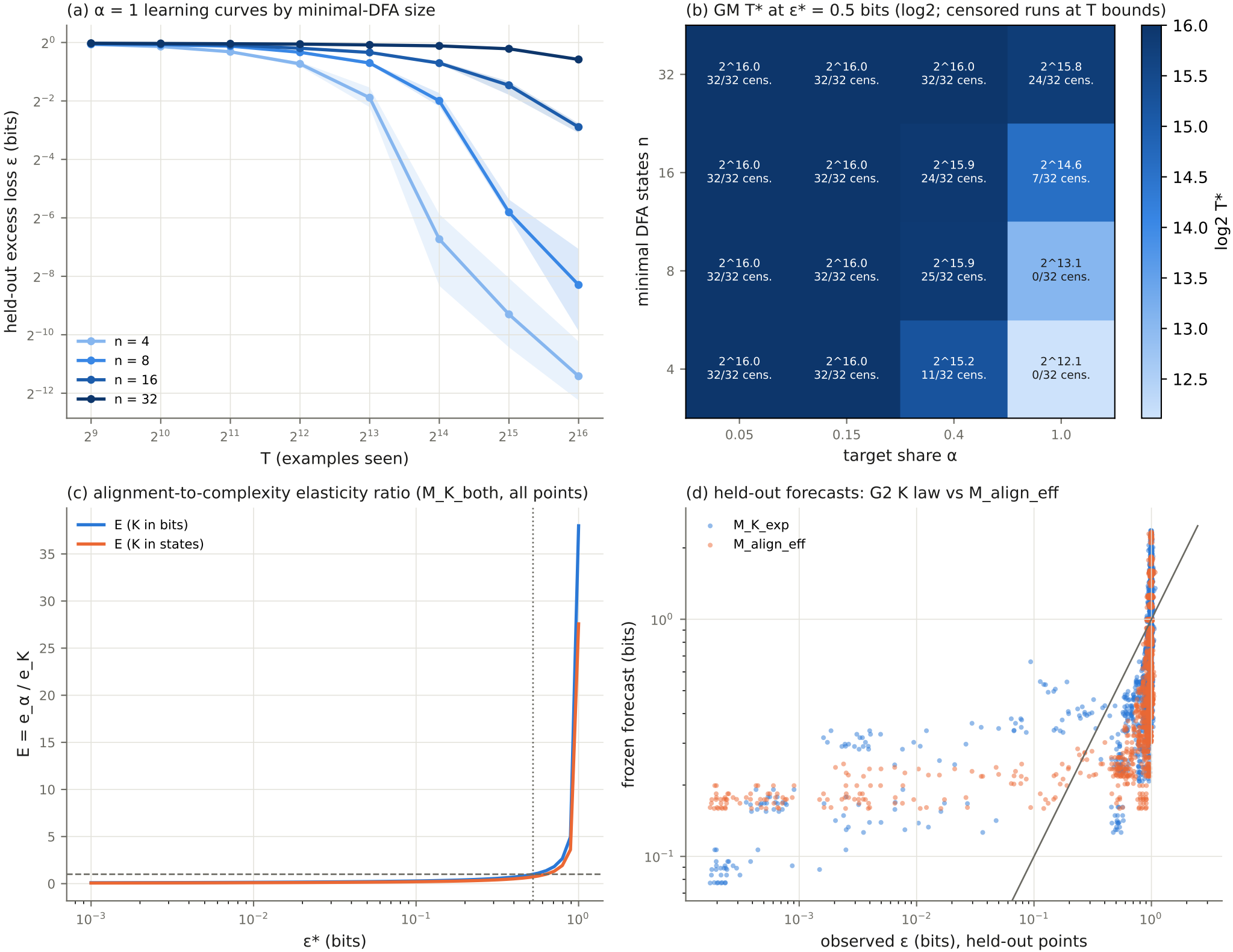

(a) α = 1 learning curves by minimal-DFA size
held-out excess loss ε (bits)
T (examples seen)
n = 4
n = 8
n = 16
n = 32
(b) GM T* at ε* = 0.5 bits (log2; censored runs at T bounds)
minimal DFA states n
target share α
log2 T*
2^16.0 32/32 cens.
2^16.0 32/32 cens.
2^16.0 32/32 cens.
2^15.8 24/32 cens.
2^16.0 32/32 cens.
2^16.0 32/32 cens.
2^15.9 24/32 cens.
2^14.6 7/32 cens.
2^16.0 32/32 cens.
2^16.0 32/32 cens.
2^15.9 25/32 cens.
2^13.1 0/32 cens.
2^16.0 32/32 cens.
2^16.0 32/32 cens.
2^15.2 11/32 cens.
2^12.1 0/32 cens.
(c) alignment-to-complexity elasticity ratio (M_K_both, all points)
E (K in bits)
E (K in states)
E = e_α / e_K
ε* (bits)
(d) held-out forecasts: G2 K law vs M_align_eff
M_K_exp
M_align_eff
frozen forecast (bits)
observed ε (bits), held-out points